

Anderson localisation with self-avoiding-walks

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1 Introduction

The Anderson localization model was introduced by the physicist P.W. Anderson in 1958 to explain the quantum mechanical effects of disorder in materials. Anderson discusses the behaviour of electrons in dirty crystals, which is the quantum mechanical analogue of a random walk in a random environment. The model considers a d -dimensional grid, in which each site represents an atom. From an heuristical point of view, we also consider electrons that jump from atom to atom and are subject to an external random potential, which influences their motion in the material. When these particles cannot jump, it indicates the absence of quantum transport. This phenomenon is called Anderson localisation. This model deals with conditions for which the electrons stay trapped in a specific region and don't generate electric conduction, defining an insulator. This is in sharp contrast to the behaviour in ideal crystals, which are always conductors.

Anderson's work, and that of N.F.Mott and J.H. van Vleck, won the 1977 physics Nobel prize for "their fundamental theoretical investigations of the electronic structure of magnetic and disordered systems". Since then, the study of random operators has become an important field of mathematical physics, which has led to a tremendous amount of research activity and many mathematical results.

The two important methods to prove Anderson localization are the *multiscale analysis* (MSA), developed by Froehlich and Spencer [1] in 1983, and the *fractional moment method* (FMM) by Aizenman and Molchanov in 1993. The method of multiscale analysis, by now, has lead to a multitude of results on spectral and dynamical localization for a wide range of models. In this thesis we will focus on the fractional moment method, which has provided a simple perspective on localization and has enabled exponentially decaying bounds on expectation values of various propagation kernels.

Structure of the thesis In the first Chapter we are going to introduce the Anderson model, in particular the properties of its components, of its spectrum and the main results known to date. In the second Chapter, we are going to discuss spectral and dynamical localisation. Moreover, we will state also the exponential dynamical localisation and how it implies that all moments of the position operator are bounded in time. In Chapter 3 we introduce the Green function as the resolvent of the model and see how its fractal power's exponential decay leads to localisation. Specifically we are

going to use simple tools such as the resolvent identity of linear operators and the a priori bound as a consequence. In Chapter 4 we give the preliminary definitions for self-avoiding-walks methods and obtain the main results of this thesis using a new representation for the Green function. The last Chapter offers a comparison between the standard model with one-site jumps and the model with long-range jumps.

Prerequisites It is useful to have some familiarity with measure theory, basic probabilistic concepts such as independence and continuous distributions, foundations of the theory of linear operators in Hilbert spaces, up to a general idea of the consequences of the spectral theorem for self adjoint operators.

1.1 The Anderson Model

Transport phenomena in a disordered material are often described via discrete random Schrödinger operator. In quantum mechanics, it corresponds to the classical total energy decomposing into kinetic and potential energy. On the lattice $\mathbb{Z}^d$, it takes the form of an infinite random matrix called the *Hamiltonian*:

$$H_\omega = -\Delta + \lambda V_\omega.$$

where $-\Delta$ is a deterministic matrix and λV_ω is a diagonal matrix where V_ω has random entries and λ is the disorder parameter.

This operator acts on the Hilbert space

$$\ell^2(\mathbb{Z}^d) = \{u : \mathbb{Z}^d \rightarrow \mathbb{C} : \sum_{x \in \mathbb{Z}^d} |u(x)|^2 < \infty\}$$

with inner product $\langle u, v \rangle = \sum_n \overline{u(x)}v(x)$.

In quantum mechanics, a physical state describing a single particle (such as an electron) is represented by a normalized wave function $\psi \in \ell^2(\mathbb{Z}^d)$ satisfying

$$\|\psi\|_{\ell^2}^2 = \sum_{x \in \mathbb{Z}^d} |\psi(x)|^2 = 1. \quad (1.1)$$

The quantity $P(x) = |\psi(x)|^2$ defines a probability mass function on the lattice $\mathbb{Z}^d$: it represents the probability of finding the particle at site $x \in \mathbb{Z}^d$. This functions are the solutions to the time-evolving Schrödinger operator $H_\omega \phi(t) = i\partial_t \phi(t)$. Because the time evolution operator e^{-itH_ω} is unitary on $\ell^2(\mathbb{Z}^d)$, the total probability is strictly conserved at all times $t \in \mathbb{R}$:

$$\|\psi(t)\|_{\ell^2}^2 = \|e^{-itH_\omega} \psi_0\|_{\ell^2}^2 = \|\psi_0\|_{\ell^2}^2 = 1. \quad (1.2)$$

In particular, if the particle is initially placed at a specific site $y \in \mathbb{Z}^d$ with $\psi(0) = \delta_y$, the transition probability to find the particle at site $x \in \mathbb{Z}^d$ at time t is given by the squared transition amplitude:

$$P(y \rightarrow x; t) = |\langle \delta_x, e^{-itH_\omega} \delta_y \rangle|^2. \quad (1.3)$$

As we'll see later on, under certain localisation conditions, P tends to 0 faster than any polynomial at all t for $|x - y| \rightarrow \infty$.

1.1.1 The Hamiltonian

The Descrete Laplacian is the descrete analogue of the usual negative Laplacian $(-\sum_j \partial^2/\partial x_j^2)$. For the functions $u \in \ell^2(\mathbb{Z}^d)$ it is defined as:

$$-\Delta u(x) := \sum_{k \in \mathbb{Z}^d, |x-k|=1} u(x) - u(k) \quad \text{for } x \in \mathbb{Z}^d. \quad (1.4)$$

where $|\cdot|$ denotes the ℓ^2 norm.

The discrete Laplacian represents the kinetic term and connects neighbour sites on the lattice, such that the electron can transition from one to another.

For each site x , there are $2d$ possible k such that $|x - k| = 1$. Thus, we can define an infinte matrix such that:

$$(-\Delta)_{x,k} = \begin{cases} 2d & \text{if } x = k \quad (\text{on the diagonal}) \\ -1 & \text{if } |x - k| = 1 \\ 0 & \text{otherwise} \end{cases}$$

For $u \in \ell^2(\mathbb{Z}^d)$, $-\Delta u$ is summable in ℓ^2 , bounded, and self-adjoint.

The first claim (summability) is shown by:

$$\begin{aligned} \sum_{x \in \mathbb{Z}^d} |\Delta u(x)|^2 &= \sum_{x \in \mathbb{Z}^d} \left| \sum_{k \in \mathbb{Z}^d, |x-k|=1} (u(x) - u(k)) \right|^2 \\ &\leq 2d \sum_{x \in \mathbb{Z}^d} \left(\sum_{|x-k|=1} |u(x) - u(k)|^2 \right) \\ &\leq 2d \sum_{x \in \mathbb{Z}^d} \left(2 \sum_{|x-k|=1} (|u(x)|^2 + |u(k)|^2) \right) \\ &\leq 2d \cdot 2d \cdot 4 \|u\|_2^2 < \infty \end{aligned}$$

where we used the bound over the number of neighbors, triangle inequality, Cauchy Schwarz and the invariance under translation of the ℓ^2 norm.

The boundedness comes from:

$$\frac{\|\Delta u\|_2^2}{\|u\|_2^2} \leq 16d^2 \implies \|\Delta\|_2 \leq 4d.$$

To prove self-adjointness, we only need to show symmetry (given $-\Delta$ is bounded). For $u, v \in \ell^2(\mathbb{Z}^d)$ and $k \in \mathbb{Z}^d$:

$$\begin{aligned}
\langle u, -\Delta v \rangle &= \sum_{x \in \mathbb{Z}^d} \overline{u(x)} (-\Delta v(x)) = \sum_{x \in \mathbb{Z}^d} \overline{u(x)} \left(\sum_{|x-k|=1} v(x) - v(k) \right) \\
&= \sum_{x \in \mathbb{Z}^d} \left(\overline{u(x)} 2dv(x) - \sum_{|x-k|=1} \overline{u(x)} v(k) \right) = \sum_{x \in \mathbb{Z}^d} 2d\overline{u(x)} v(x) - \sum_{x \in \mathbb{Z}^d} \sum_{|x-k|=1} \overline{u(k)} v(x) \\
&= \sum_{x \in \mathbb{Z}^d} v(x) \left(\sum_{|x-k|=1} \overline{u(x)} - \overline{u(k)} \right) = \sum_{x \in \mathbb{Z}^d} v(x) \overline{(-\Delta u(x))} = \langle -\Delta u, v \rangle.
\end{aligned}$$

The spectrum of a bounded, self-adjoint operator is real-valued, and in this case equal to the closed interval $[0, 4d]$. In this range, we find the levels of kinetic energy that an electron owns. In order to understand this result better, let's consider the discrete Fourier transform:

$$\mathcal{F}: \ell^2(\mathbb{Z}^d) \longrightarrow L^2([0, 2\pi)^d)$$

such that:

$$(\mathcal{F}u)(x) := \lim_{N \rightarrow \infty} (2\pi)^{-\frac{d}{2}} \sum_{|n| \leq N} u(n) e^{-ix \cdot n}, \quad x \in [0, 2\pi)^d.$$

We consider now calculations for $u \in \ell^1(\mathbb{Z}^d)$. For ℓ^1 is a dense subset of ℓ^2 and $\mathcal{F}$ unitary, we can extend the results by continuity for all $u \in \ell^2(\mathbb{Z}^d)$

$$\begin{aligned}
(\mathcal{F}(-\Delta u))(x) &= (2\pi)^{-\frac{d}{2}} \sum_{n \in \mathbb{Z}^d} (-\Delta u)(n) e^{-ix \cdot n} \\
&= (2\pi)^{-\frac{d}{2}} \sum_{n \in \mathbb{Z}^d} \left(\sum_{|k-n|=1} (u(n) - u(k)) \right) e^{-ix \cdot n} \\
&= (2\pi)^{-\frac{d}{2}} \sum_{n \in \mathbb{Z}^d} \left(2d u(n) - \sum_{|k-n|=1} u(k) \right) e^{-ix \cdot n} \\
&= 2d (\mathcal{F}u)(x) - (2\pi)^{-\frac{d}{2}} \sum_{n \in \mathbb{Z}^d} \sum_{|k-n|=1} u(k) e^{-ix \cdot n} \\
&= 2d (\mathcal{F}u)(x) - (2\pi)^{-\frac{d}{2}} \sum_{k \in \mathbb{Z}^d} u(k) \sum_{|m|=1} e^{-ix \cdot (k+m)} \quad (\text{for } m = n - k) \\
&= 2d (\mathcal{F}u)(x) - \left((2\pi)^{-\frac{d}{2}} \sum_{k \in \mathbb{Z}^d} u(k) e^{-ix \cdot k} \right) \sum_{|m|=1} e^{-ix \cdot m} \\
&= 2d (\mathcal{F}u)(x) - (\mathcal{F}u)(x) \sum_{j=1}^d (e^{-ix_j} + e^{ix_j}) \\
&= \left(2 \sum_{j=1}^d (1 - \cos(x_j)) \right) (\mathcal{F}u)(x).
\end{aligned}$$

We obtain the multiplication operator:

$$\mathcal{F} \Delta \mathcal{F}^{-1} = 2 \sum_{j=1}^d (1 - \cos(x_j))$$

on $L^2([0, 2\pi)^d)$ in the variable $x = (x_1, \dots, x_d)$. We notice that the Laplacian doesn't have eigenvalues in $\ell^2(\mathbb{Z}^d)$. Another similarity of Δ with the continuum Laplacian is that it has plane waves as generalized eigenfunctions. To see this, let $x \in [0, 2\pi)^d$ and set

$$\phi_x(n) := e^{-in \cdot x}$$

While $\phi_x \notin \ell^2(\mathbb{Z}^d)$, Δ acts on it as

$$\begin{aligned}
(\Delta\phi_x)(n) &= \sum_{|k|=1} (e^{-in \cdot x} - e^{-i(n+k) \cdot x}) \\
&= 2d\phi_x(n) - \left(\sum_{j=1}^d 2\cos(x_j) \right) \phi_x(n) = \left(2 \sum_{j=1}^d (1 - \cos(x_j)) \right) \phi_x.
\end{aligned}$$

Thus ϕ_x is a bounded generalized normalized eigenfunction of Δ to the spectral value $2 \sum_{j=1}^d (1 - \cos(x_j))$. thus, the spectrum is continuous and bounded in $[0, 2(2d)]$.

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The Random Potential is expressed as the multiplication operator $V_\omega(x) = \omega_x$. It is a diagonal matrix, where $\omega = (\omega_x)_{x \in \mathbb{Z}^d}$ is the set of the independent identically distributed real-valued random entries. Physically, V_ω represents the random electric potential created by nuclei at the sites $n \in \mathbb{Z}^d$.

Definition 1.1 (i.i.d. random variables). $(\omega_n)_{n \in \mathbb{Z}^d}$ are **identically distributed** if there exists a Borel probability measure μ on $\mathbb{R}$ such that $\mathbb{P}(\omega_n \in A) = \mu(A)$ for all $n \in \mathbb{Z}^d$ and Borel sets $A \subset \mathbb{R}$. $(\omega_n)_{n \in \mathbb{Z}^d}$ are **independent** if

$$\mathbb{P}(\omega_{n_1} \in A_1, \dots, \omega_{n_l} \in A_l) = \prod_{k=1}^l \mathbb{P}(\omega_{n_k} \in A_k) = \prod_{k=1}^l \mu(A_k)$$

for each finite subset $\{n_1, \dots, n_l\}$ of $\mathbb{Z}^d$ and arbitrary Borel sets $A_1, \dots, A_l \subset \mathbb{R}$.

In this thesis we assume that ω_n are uniformly distributed in the real closed interval $[-1, 1]$. This means that $d\mu(\omega_n) = \rho(\omega_n)d\omega_n$ where the continuous density is given by $\rho = \frac{1}{2}\chi_{[-1, 1]}$. It is possible to generalise the results to all absolutely continuous distributions where ρ has compact bounded support and to the Gaussian distribution.

As a diagonal operator, the spectrum of V_ω consists of the random entries ω_n , each corresponding to an eigenvector given by the Dirac delta $\delta_n \in \ell^2(\mathbb{Z}^d)$.

As for the discrete laplacian, the random term is summable in $\ell^2(\mathbb{Z}^d)$, bounded and self-adjoint. Self-adjointness of the maximal multiplication operator by a real-valued function is a fact. For definition V_ω is bounded in the closed interval $[-1, 1] \subset \mathbb{R}$, and since $\|V_\omega u\|_{\ell^2}^2 \leq \|u\|_{\ell^2}^2$ it is also summable in $\ell^2(\mathbb{Z}^d)$.

The Disorder Parameter is represented by $\lambda > 0$. The value of λ is a great indicator for localisation, which makes it one of the central objects of interest of this thesis. Let's consider the two extreme cases:

- For $\lambda = 0$ the Hamiltonian loses the random term and only considers the Laplacian. Without disorder, the material behaves like a perfect crystal without localisation. The state functions are continuous plane waves and the spectrum is also continuous between $[0, 4d]$.
- For $\lambda \gg 1$ the kinetic hopping term becomes negligible compared to the random potential, so that $H_\omega \approx \lambda V_\omega$. In this scenario electrons are unable to hop between adjacent sites, and the material behaves as an insulator. In this limit, the system exhibits trivial localization: the eigenfunctions are localized Dirac deltas δ_x , and the spectrum is confined to the compact interval $[-\lambda, \lambda]$

1.1.2 The spectrum

The next goal is to determine the spectrum of H_ω . Because the random potential entries $(\omega_x)_{x \in \mathbb{Z}^d}$ are independent and identically distributed, the random Hamiltonian $H_\omega = -\Delta + \lambda V_\omega$ is an ergodic family of bounded self-adjoint operators. A standard consequence of ergodic operator theory is that the spectrum of H_ω is almost surely deterministic (see, for example, [Pastur1980; Stolz2011]): there exists a closed, non-random set $\Sigma \subset \mathbb{R}$ such that

$$\sigma(H_\omega) = \Sigma \quad \text{for } \mathbb{P}\text{-almost all } \omega \in \Omega. \quad (1.5)$$

More specifically, as proven in [Theorem 2 GUNTERSTOLZ], the almost sure spectrum can be determined explicitly as the Minkowski sum of the spectrum of the kinetic energy operator and the support of the random potential distribution:

$$\Sigma = \sigma(-\Delta) + \lambda \text{supp}(\mu). \quad (1.6)$$

Recall from Section 1.1.1 that under our non-negative convention for the discrete Laplacian, its spectrum is the band $\sigma(-\Delta) = [0, 4d]$. The random potential V_ω has i.i.d. entries distributed uniformly on $[-1, 1]$, making its almost sure spectrum the interval $[-\lambda, \lambda]$. Consequently, the almost sure spectrum of H_ω is given by the compact interval

$$\sigma(H_\omega) = [0, 4d] + [-\lambda, \lambda] = [-\lambda, 4d + \lambda]. \quad (1.7)$$

This confirms that the spectrum is a bounded subset of $\mathbb{R}$.

Before characterising the physical meaning of this spectrum, we need some spectral theory preliminaries. Let μ be a Borel measure on $\mathbb{R}$. Consider now $P = \{x \in \mathbb{R} \mid \mu(\{x\}) \neq 0\}$ the set of **pure points** of μ . We define the pure point measure as

$$\mu_{pp}(X) = \mu(P \cap X)$$

for $X \subset \mathbb{R}$. In particular, there exists a continuous measure μ_c such that $\mu_c(\{x\}) = 0$ for all $x \in X$ and $\mu_c = \mu - \mu_{pp}$. Since $\mu_{pp}(P^c) = 0$ and $\mu_c(P) = 0$, the two measures are concentrated on disjoint sets, and thus mutually singular.

One can apply this same division also for spectral measures:

Definition 1.2. Let $A : \mathcal{H} \rightarrow \mathcal{H}$ be a bounded, self-adjoint operator. Let $\phi \in \mathcal{H}$ and $f \in C(\sigma(A))$. Then there is a unique measure μ_ϕ on the compact set $\sigma(A)$ with $\langle \phi, f(A)\phi \rangle = \int_{\sigma(A)} f(\lambda) d\mu_\phi$. The measure μ_ϕ is called the *spectral measure associated with the vector ϕ* .

Thus, by the spectral theorem for self-adjoint operators ([RUDIN or REED]), the Hilbert space (for example also $\ell^2(\mathbb{Z}^d)$) decomposes uniquely into an orthogonal direct sum of invariant subspaces:

$$\mathcal{H} = \mathcal{H}_{pp} \oplus \mathcal{H}_c, \quad (1.8)$$

where:

- $\mathcal{H}_{pp} = \{\phi \mid \mu_\phi \text{ is pure point}\};$
- $\mathcal{H}_c = \{\phi \mid \mu_\phi \text{ is continuous}\}$

In the case of self-adjoint bounded operators, the restriction to these subspaces defines the components of the spectrum:

$$\sigma(H_\omega) = \sigma_{pp}(H_\omega) \cup \sigma_c(H_\omega) \quad (1.9)$$

where $\sigma_{\text{pp}}(H_\omega) = \overline{\{\lambda \in \mathbb{R} \mid \lambda \text{ is an eigenvalue of } H_\omega\}}$. These sets need not to be disjoint, however.

From a physical perspective, states in $\mathcal{H}_{\text{pp}}$ correspond to bound states that remain confined to finite spatial regions for all times, whereas states in the continuous subspace $\mathcal{H}_{\text{c}}$ represent scattering states associated with transport and wave-packet spreading. This heuristic meaning is mathematically expressed by the RAGE theorem (named after Ruelle, Amrein, Georgescu, and Enss):

Theorem 1.3 (RAGE Theorem [Cycon1987; HunzikerSigal2000]). *Let H be a self-adjoint Schrödinger operator on $\ell^2(\mathbb{Z}^d)$. Then:*

(i) $\phi \in \mathcal{H}_{\text{pp}}$ if and only if

$$\lim_{R \rightarrow \infty} \sup_{t \in \mathbb{R}} \|\chi_{\{|x| > R\}} e^{-itH} \phi\| = 0. \quad (1.10)$$

(ii) $\psi \in \mathcal{H}_{\text{c}}$ if and only if for all $R > 0$,

$$\lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T \|\chi_{\{|x| \leq R\}} e^{-itH} \psi\|^2 dt = 0. \quad (1.11)$$

Here, $\chi_{\{|x| \leq R\}}$ and $\chi_{\{|x| > R\}}$ denote the projection operators corresponding to the characteristic functions of the ball $\{x \in \mathbb{Z}^d : |x| \leq R\}$ and its complement, respectively.

In terms of this classification, Anderson localization corresponds to the complete suppression of continuous spectrum, ensuring that $\sigma(H_\omega) = \sigma_{\text{pp}}(H_\omega)$ almost surely.

1.2 Main results and Open Problems

There exist many studies on the Anderson localisation model, led by both physicist and mathematicians. Various discoveries have been made in the past years and yet deep challenges remain.

The main result of this thesis states that if $s \in (0, 1)$ and $\lambda > \lambda_0(s)$

$$\mathbb{E} \left(\left| \langle \delta_x, \frac{1}{(H_\omega - z)} \delta_y \rangle \right|^s \right) \leq C e^{\mu|x-y|} \quad (1.12)$$

uniformly in $z \in \mathbb{C} \setminus \mathbb{R}$ and for all $x, y \in \mathbb{Z}^d$, where λ_{And} is a fixed value proposed by P.W. Anderson, and rigorously proven later on by J. Schenker

|| using self-avoiding walks. Finally we prove that localisation follows from this bound.

As we can see, the resolvent is well-defined only if z is not an element of $\sigma(H_\omega)$. In those cases, the resolvent blows up. The reason why the *fractional moments method* was so revolutionary, is that by avereging the fractional power of the resolvent, we avoid the blow ups for $z \rightarrow \sigma(H_\omega)$ and the bound holds uniformly. Trivially we obtain:

$$\int_{\Sigma} \frac{1}{|x|^s} dx = \left[\frac{x^{1-s}}{1-s} \right]_{\Sigma} < \infty \quad \text{for } s \in (0, 1)$$

where $\Sigma \subset \mathbb{R}$. Originally, absence of conductivity was proved using the squared-power instead of the fractional, which creates problems at zero.

Very important results concern the value of λ . This are the known results and open problems.

- $d = 1$.

For the one-dimensional grid, it has been proven, that localisation holds for all $\lambda > 0$.

- $d \geq 2$.

For $d = 2$ it is expected that the Hamiltonian shows pure point spectrum for very small disorder chaneges, but no mathematical proof has been concluded. For $d \geq 3$, to find the exact value of λ , until which the material is still a conductor, is an open problem. The best that we can prove is given in this thesis and only states, that localisation shows from λ_{And} on.

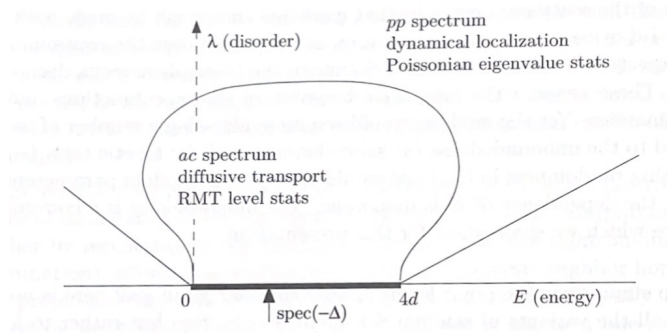


Figure 1: For small disorder localisation is not expected

2 Localisation properties

We understood what localisation means from an heuristical point of view. Mathematically, it can be described from spectral and dynamical properties.

2.1 Spectral Localisation

Definition 2.1 (Spectral Localisation). random Schrödinger operator $H = H_\omega$ exhibits **spectral localisation** if H_ω almost surely has only pure point spectrum, that is,

$$\sigma(H_\omega) = \sigma_{pp}(H_\omega) \quad \text{and} \quad \sigma_c(H_\omega) = \emptyset \quad \text{almost surely.} \quad (2.1)$$

2.1.1 Absence of transport

$$\langle x^2 \rangle_{P_{H_\omega \in [a,b]} \phi_0}(t) = \| |x| e^{-itH_\omega} P_{H_\omega \in [a,b]} \phi_0 \|^2 \quad (5)$$

If the electrons with energies in $[a, b]$ move ballistically with average velocity v_{av} , then roughly $x(t) \sim v_{av}t$ for large times. Hence $\langle x^2 \rangle_{P_{H_\omega \in [a,b]} \phi_0}(t)$ will be proportional to t^2 for large t .

On the other hand, if the electrons in the energy interval $[a, b]$ are localized, then it should be natural to expect that $\langle x^2 \rangle_{P_{H_\omega \in [a,b]} \phi_0}(t)$ is bounded uniformly in t . It is known (Simon, 1990) that pure point spectrum implies absence of ballistic motion,

$$\lim_{t \rightarrow \infty} \frac{\langle x^2 \rangle_{P_{H_\omega \in [a,b]} \phi_0}(t)}{t^2} = 0$$

for all compactly supported initial conditions ϕ_0 , as soon as the spectrum in $[a, b]$ is pure point. However, del Rio et al. (1996) constructed examples of (non-random) one-dimensional Schrödinger operators with pure point spectrum for all energies and exponentially localized eigenfunctions for which

$$\limsup_{t \rightarrow \pm\infty} \frac{\langle x^2 \rangle_{\phi_0}(t)}{t^{2-\delta}} = \infty \quad \text{for all } \delta > 0, \quad (6)$$

for a large class of compactly supported initial conditions ϕ_0 . *Il fallimento risiede nella libertà delle costanti $C_{\omega,n}$: se $C_{\omega,n}$ cresce incontrollatamente con n , le code delle autofunzioni possono sovrapporsi su scale spaziali arbitrariamente grandi.*

2.2 Dynamical Localisation

Definition 2.2 (Dynamical localisation).

$$\sup_{t \in \mathbb{R}} \| |X|^p e^{-itH_\omega} \psi \| < \infty \quad \text{almost surely,} \quad (2.2)$$

where the position operator $|X|$ is defined by $(|X|\phi)(n) = |n|\phi(n)$ and for all φ with compact support.

Definition 2.3 (Exponential localisation).

$$\mathbb{E} \left(\sup_{t \in \mathbb{R}} |\langle e_j, e^{-itH_\omega} e_k \rangle| \right) \leq C e^{-\mu|j-k|} \quad (2.3)$$

Remark 2.4. Dynamical localization in the form (2.3) is a strong form of asserting that solutions of the time-dependent Schrödinger equation $H_\omega \phi(t) = i\partial \phi(t)$ are staying localized in space, uniformly for all times, and thus shows the absence of quantum transport.

Proposition 2.5. *Exponential localization implies that all moments of the position operator are bounded in time for almost every realisation of V_ω , i.e., dynamical localisation.*

Proof. To see how (2.2) follows from (2.3), we assume that $\psi(k) = 0$ for $|k| > R$. Then

$$\begin{aligned} \| |X|^p e^{-itH_\omega} \psi \|^2 &= \sum_j |\langle e_j, |X|^p e^{-itH_\omega} \psi \rangle|^2 \\ &= \sum_j |j|^{2p} \left| \sum_{|k| \leq R} \langle e_j, e^{-itH_\omega} e_k \rangle \psi(k) \right|^2 \\ &\leq \sum_j \sum_{|k| \leq R} |j|^{2p} |\langle e_j, e^{-itH_\omega} e_k \rangle|^2 \|\psi\|^2, \end{aligned}$$

where the last step used the Cauchy-Schwarz inequality. We can drop the square from $|\langle e_j, e^{-itH_\omega} e_k \rangle|^2$ (as this number is bounded by 1) and then take

expectations to get

$$\begin{aligned}
\mathbb{E} \left(\sup_t \| |X|^p e^{-itH_\omega} \psi \|^2 \right) &\leq \sum_j \sum_{|k| \leq R} |j|^{2p} \mathbb{E} \left(\sup_t |\langle e_j, e^{-itH_\omega} e_k \rangle| \right) \|\psi\|^2 \\
&\leq C \sum_j \sum_{|k| \leq R} |j|^{2p} e^{-\mu|j-k|} \|\psi\|^2 \\
&< \infty.
\end{aligned}$$

□

Remark 2.6. The reason we obtain dynamical localization *almost surely* (rather than for every single realization of the potential) is that our bounds apply to the expected value over the disorder, rather than directly to the norm. As shown in the following standard lemma, bounding the expectation of a non-negative quantity guarantees it is finite with probability one.

Lemma 2.7. *Let $(\Omega, \Sigma, \mathbb{P})$ be a probability space and let $A : \Omega \rightarrow [0, \infty]$ be a nonnegative, Σ -measurable random variable. Define $\Omega_0 := \{\omega \in \Omega \mid A(\omega) = \infty\}$. If $\mathbb{E}[A] < \infty$, then $\mathbb{P}(\Omega_0) = 0$; equivalently, $A < \infty$ $\mathbb{P}$ -almost surely.*

Proof. For every $M > 0$, Markov's inequality gives $\mathbb{P}(A \geq M) \leq \frac{\mathbb{E}[A]}{M}$. Since $\Omega_0 \subseteq \{A \geq M\}$ for every M , we get $\mathbb{P}(\Omega_0) \leq \frac{\mathbb{E}[A]}{M} \rightarrow 0$ as $M \rightarrow \infty$, so $\mathbb{P}(\Omega_0) = 0$. □

2.3 From dynamical to spectral localisation

Proposition 2.8. *Suppose that dynamical localization in the form*

$$\mathbb{E} \left(\sup_{t \in \mathbb{R}} |\langle e_j, e^{-it h_\omega} \chi_I(h_\omega) e_k \rangle| \right) \leq C e^{-\mu|j-k|} \quad (2.4)$$

holds in an open interval I . Then h_ω almost surely has pure point spectrum in I .

Proof. For a discrete Schrödinger operator $h = h_0 + V$ in $\ell^2(\mathbb{Z}^d)$ let $P_{\text{cont}}(h)$ be the projection onto its continuous spectral subspace. Then the RAGE-Theorem says that for every $\psi \in \ell^2(\mathbb{Z}^d)$,

$$\|P_{\text{cont}}(h) \chi_I(h) \psi\|^2 = \lim_{R \rightarrow \infty} \lim_{T \rightarrow \infty} \int_0^T \frac{dt}{T} \|\chi_{\{|x| \geq R\}} e^{-it h} \chi_I(h) \psi\|^2. \quad (2.5)$$

If ψ has finite support, say $\text{supp } \psi \subset \{|x| \leq r\}$, then

$$\begin{aligned} \|\chi_{\{|x| \geq R\}} e^{-ith} \chi_I(h) \psi\|^2 &\leq \|\chi_{\{|x| \geq R\}} e^{-ith} \chi_I(h) \chi_{\{|x| \leq r\}}\|^2 \|\psi\|^2 \\ &\leq \|\chi_{\{|x| \geq R\}} e^{-ith} \chi_I(h) \chi_{\{|x| \leq r\}}\| \|\psi\|^2 \\ &\leq \sum_{|x| \geq R, |y| \leq r} |\langle e_x, e^{-ith} \chi_I(h) e_y \rangle| \|\psi\|^2, \end{aligned}$$

where dropping a square is allowed as $\|\chi_{\{|x| \geq R\}} e^{-ith} \chi_I(h) \chi_{\{|x| \leq r\}}\| \leq 1$.

Taking expectations in (2) implies, after using Fatou and Fubini,

$$\mathbb{E} (\|P_{\text{cont}}(h_\omega) \chi_I(h_\omega) \psi\|^2) \leq \lim_{R \rightarrow \infty, T \rightarrow \infty} \int_0^T \frac{dt}{T} \sum_{|x| \geq R, |y| \leq r} \mathbb{E} \left(|\langle e_x, e^{-ith_\omega} \chi_I(h_\omega) e_y \rangle| \right) \|\psi\|^2. \quad (2.6)$$

By (2.4) we have $\mathbb{E} (|\langle e_x, e^{-ith_\omega} \chi_I(h_\omega) e_y \rangle|) \leq C e^{-\mu|x-y|}$ uniformly in t , which bounds the right hand side of (2.6) by

$$\leq \lim_{R \rightarrow \infty} \tilde{C} \sum_{|x| \geq R, |y| \leq r} e^{-\mu|x-y|} = 0.$$

We conclude that $P_{\text{cont}}(h_\omega) \chi_I(h_\omega) \psi = 0$ for almost every ω and every ψ of finite support. The latter are dense in $\ell^2(\mathbb{Z}^d)$ and thus $P_{\text{cont}}(h_\omega) \chi_I(h_\omega) = 0$ almost surely, meaning that the spectrum in I is pure point. $\square$

3 Fractional moments

In this section we are going to see the important lemmas and theorems that help us proving localisation. In particular we see the power of the fractional moment method.

3.1 The Green's function

The *resolvent operator* is defined as:

$$G_z(x, y) := \langle \delta_x, (H_\omega - z)^{-1} \delta_y \rangle \quad (3.1)$$

for all $x, y \in \mathbb{Z}^d$ and is well-defined for $z \in \mathbb{C} \setminus \sigma(H_\omega)$

3.2 Localisation at large disorder

Theorem 3.1 (Aizenman–Molchanov, 1993). *Let $0 < s < 1$. Then there exists $\lambda_0 > 0$ such that for $\lambda \geq \lambda_0$ there are $C < \infty$ and $\mu > 0$ with*

$$\mathbb{E}(|G_z(x, y)|^s) \leq C e^{-\mu|x-y|} \quad (3.2)$$

uniformly in $x, y \in \mathbb{Z}^d$ and $z \in \mathbb{C} \setminus \mathbb{R}$. In particular, if λ_{And} satisfies $\lambda_{\text{And}} = \mu_d e \ln \lambda_{\text{And}}$ then for all $\lambda > \lambda_{\text{And}}, \lambda > e$ and $\epsilon > 0$ we have

$$\mathbb{E}(|G_z(x, y)|^{1 - \frac{1}{\ln \lambda}}) \leq \beta(\lambda) K_\epsilon \ln \lambda e^{-m_\epsilon(\lambda)|x-y|} \quad (3.3)$$

where

$$m_\epsilon(\lambda) = -\ln \beta(\lambda) - \ln(\mu_d + \epsilon) > 0.$$

Lemma 3.2 (A priori bound). *There exists a constant $C_1 = C_1(s, \rho) < \infty$ such that*

$$\mathbb{E}_x(|G_z(x, x)|^s) \leq C_1 \lambda^{-s} \quad (3.4)$$

for all $x \in \mathbb{Z}^d$, $z \in \mathbb{C} \setminus \mathbb{R}$, and $\lambda > 0$. Here

$$C_1 = \frac{1}{1-s}$$

for $\omega_x \sim \mathcal{U}[-1, 1]$

Lemma 3.3 (Resolvent identity). *For a linear operator $A : \mathcal{D}(A) \rightarrow \mathcal{H}$ and $z_1, z_2 \in \varrho(A)$,*

$$(A - z_1)^{-1} - (A - z_2)^{-1} = (A - z_1)^{-1}(z_1 - z_2)(A - z_2)^{-1}$$

and moreover, if $B : \mathcal{D}(B) \rightarrow \mathcal{H}$ is defined on a common domain, $\mathcal{D}(A) = \mathcal{D}(B)$,

$$(A - z)^{-1} - (B - z)^{-1} = (A - z)^{-1}(B - A)(B - z)^{-1}$$

for any $z \in \varrho(A) \cap \varrho(B)$.

Proof of a priori bound. This proof demonstrates the simplicity of the fundamental idea underlying the FMM. For fixed $x \in \mathbb{Z}^d$, write $\omega = (\hat{\omega}, \omega_x)$ where $\hat{\omega}$ is short for $(\omega_u)_{u \in \mathbb{Z}^d \setminus \{x\}}$. With $P_{e_x} := \langle e_x, \cdot \rangle e_x$, the orthogonal projection onto the span of e_x , we can separate the ω_x and $\hat{\omega}$ dependence of H_ω as

$$H_\omega = -\Delta + \lambda V_{\hat{\omega}} + \lambda \omega_x P_{e_x} = H_{\hat{\omega}} + \lambda \omega_x P_{e_x}.$$

For the operators $(H_\omega - z)$ and $(H_{\hat{\omega}} - z)$, the resolvent identity yields

$$(H_\omega - z)^{-1} = (H_{\hat{\omega}} - z)^{-1} - \lambda \omega_x (H_{\hat{\omega}} - z)^{-1} P_{e_x} (H_\omega - z)^{-1} \quad (3.5)$$

Taking matrix-elements we conclude for the corresponding diagonal Green functions that

$$G_z(x, x) = \hat{G}_z(x, x) - \lambda \omega_x \hat{G}_z(x, x) G_z(x, x) \quad (3.6)$$

where $\hat{G}_z$ denotes the Green's function of $H_{\hat{\omega}}$. Collecting in $G_z(x, x)$:

$$G_z(x, x) = \frac{1}{a + \lambda \omega_x} \quad \text{with } a = \frac{1}{\hat{G}_z(x, x)}. \quad (3.7)$$

Note that the latter is well-defined since one can easily check that $\text{Im} \hat{G}_z(x, x) > 0$ for $\text{Im} z > 0$.

The important fact is that a is a complex number which does not depend on ω_x . Thus, writing $\mathbb{E}_x(\dots) := \int \dots \rho(\omega_x) d\omega_x$ for $\rho(\omega_x) = \frac{1}{2} \chi_{[-1, 1]}$, we find that

$$\mathbb{E}_x(|G_z(x, x)|^s) = \frac{1}{2\lambda^s} \int_{-1}^1 \frac{d\omega_x}{|\frac{a}{\lambda} + \omega_x|^s} \leq \frac{1}{2\lambda^s} \int_{-1}^1 \frac{d\omega_x}{|\omega_x|^s} = \frac{1}{(1-s)\lambda^s}, \quad (3.8)$$

with $\frac{1}{1-s}$ independent of λ and a , and thus independent of $\hat{\omega}$, z and x . □

3.2.1 From fractional moment's exponential bound to dynamical localisation

In this subsection we aim to prove the following theorem:

Theorem 3.4. *Let $I \subset \mathbb{R}$ be an open bounded interval. If there exist $s \in (0, 1)$, $C < \infty$ and $\mu > 0$ such that*

$$\mathbb{E}(|G_{E+i\eta}(x, y)|^s) \leq Ce^{-\mu|x-y|} \quad (3.9)$$

uniformly in $E \in I$ and $\eta > 0$, then dynamical localisation holds on the interval I .

Before diving in the proof, we need two helpers.

Lemma 3.5. *For a bounded and self-adjoint operator A ,*

$$\|(A - z)^{-1}\| \leq \frac{1}{\text{Im}(z)} \quad (3.10)$$

for all $z \in \mathbb{C} \setminus \mathbb{R}$.

Proof. For the definition of the operator norm, we know that

$$\|(A - z)^{-1}\| := \sup_{\mu \in \sigma((A - z)^{-1})} |\mu|$$

Let's consider a $\lambda \in \mathbb{C}$ and write $\mu = \frac{1}{\lambda - z}$. This means:

$$\begin{aligned} \frac{1}{\lambda - z} \in \sigma((A - z)^{-1}) &\iff (A - z)^{-1} - (\lambda - z)^{-1} \text{ not invertible} \\ &\iff (A - z)^{-1}(A - \lambda)(\lambda - z)^{-1} \text{ not invertible} \\ &\iff (A - \lambda) \text{ not invertible} \\ &\iff \lambda \in \sigma(A) \end{aligned}$$

Where the resolvent identity has been used in third Implication.

At this point we obtain

$$\|(A - z)^{-1}\| = \sup_{\lambda \in \sigma(A)} \frac{1}{|\lambda - z|} = \frac{1}{\inf_{\lambda \in \sigma(A)} |\lambda - z|} = \frac{1}{\text{dist}(\sigma(A), z)}$$

For a bounded and self-adjoint operator, we know that the spectrum is real-valued. For $z \in \mathbb{C} \setminus \mathbb{R}$ the minimum distance between z and $\sigma(A)$ is the imaginary component of z :

$$\|(A - z)^{-1}\| \leq \frac{1}{|\text{Im}(z)|} \quad (3.11)$$

□

This bound is one of the most important and useful in this thesis. Once again we applied the resolvent identity to find a simple yet useful result.

Proposition 3.6. *For every $s \in (0, 1)$ there exists a constant $C_1 < \infty$ only depending on s and ρ such that*

$$|\operatorname{Im} z| \mathbb{E}_x (|G_z(x, y)|^2) \leq C_1 \mathbb{E}_x (|G_z(x, y)|^s) \quad (3.12)$$

for all $z \in \mathbb{C} \setminus \mathbb{R}$ and $x, y \in \mathbb{Z}^d$.

Proof. As in the proof of 3.2 we consider the Hamiltonian

$$H^{(\alpha)} = H_\omega + \alpha P_{e_x}.$$

Its Green function will be denoted by $G^{(\alpha)}$. Applying the resolvent identity, we find

$$(H_\omega - z)^{-1} = (H^{(\alpha)} - z)^{-1} + \alpha (H_\omega - z)^{-1} P_{e_x} (H^{(\alpha)} - z)^{-1}$$

and

$$\begin{aligned} G_z(x, y) &= G_z^{(\alpha)}(x, y) + \alpha G_z(x, y) G^{(\alpha)}(x, x) \\ \Rightarrow G_z^{(\alpha)}(x, y) &= \frac{G_z(x, y)}{1 + \alpha G_z(x, x)} \\ &= \frac{1}{\alpha + G_z(x, x)^{-1}} \cdot \frac{G_z(x, y)}{G_z(x, x)}. \end{aligned} \quad (3.13)$$

For the special case $x = y$ and $\tilde{\alpha} = -\operatorname{Re} G_z(x, x)^{-1}$ we get from (3.13) and Lemma 3.5 that

$$\left| \frac{1}{\operatorname{Im} G_z(x, x)^{-1}} \right| = |G_z^{(\tilde{\alpha})}(x, x)| \leq \frac{1}{|\operatorname{Im} z|},$$

i.e. $|\operatorname{Im} G_z(x, x)^{-1}| \geq |\operatorname{Im} z|$. Inserting this into (3.13) gives

$$|\operatorname{Im} z| |G_z^{(\alpha)}(x, y)|^2 \leq \frac{|\operatorname{Im} G_z(x, x)^{-1}|}{|\alpha + G_z(x, x)^{-1}|^2} \cdot \frac{|G_z(x, y)|^2}{|G_z(x, x)|^2}. \quad (3.14)$$

On the other hand, we can bound the same expression by

$$\begin{aligned}
|\operatorname{Im} z| |G_z^{(\alpha)}(x, y)|^2 &\leq |\operatorname{Im} z| \sum_{y' \in \mathbb{Z}^d} |G_z^{(\alpha)}(x, y')|^2 \\
&= |\operatorname{Im} z| \langle e_x (H^{(\alpha)} - z)^{-1}, (H^{(\alpha)} - z)^{-1} e_x \rangle
\end{aligned}$$

For H_ω bounded and self adjoint, also $H^{(\alpha)}$ is bounded and self adjoint, da $\alpha \in \mathbb{R}$ and $\alpha < \infty$. It holds then,

$$|\operatorname{Im} z| \langle e_x (H^{(\alpha)} - z)^{-1}, (H^{(\alpha)} - z)^{-1} e_x \rangle = |\operatorname{Im} z| \langle e_x, (H^{(\alpha)} - \bar{z})^{-1} (H^{(\alpha)} - z)^{-1} e_x \rangle$$

Applying the resolvent identity:

$$(H^{(\alpha)} - \bar{z})^{-1} - (H^{(\alpha)} - z)^{-1} = (H^{(\alpha)} - \bar{z})^{-1} (\operatorname{Im}(z)) (H^{(\alpha)} - z)^{-1}$$

. We obtain:

$$\begin{aligned}
&= |\operatorname{Im} z| \langle e_x, (H^{(\alpha)} - \bar{z})^{-1} (H^{(\alpha)} - z)^{-1} e_x \rangle \\
&= |\operatorname{Im} z| \langle e_x, \frac{1}{\operatorname{Im}(z)} \left[(H^{(\alpha)} - z)^{-1} - (H^{(\alpha)} - \bar{z})^{-1} \right] e_x \rangle \\
&= |\operatorname{Im} G^{(\alpha)}(x, x; z)| \\
&= \frac{|\operatorname{Im} G_\omega(x, x; z)^{-1}|}{|\alpha + G_\omega(x, x; z)^{-1}|^2}, \tag{3.15}
\end{aligned}$$

where the last step used (3.13) with $x = y$.

For $t \geq 0$ one has $\min(1, t^2) \leq t^s$, with $s \in (0, 1)$. Using this to interpolate between (3.14) and (3.15) we get

$$|\operatorname{Im} z| |G^{(\alpha)}(x, y; z)|^2 \leq \frac{|\operatorname{Im} G_\omega(x, x; z)^{-1}|}{|\alpha + G_\omega(x, x; z)^{-1}|^2} \cdot \frac{|G_\omega(x, y; z)|^s}{|G_\omega(x, x; z)|^s}. \tag{3.16}$$

We will now use the following “re-sampling trick”, which has the effect of creating an additional random variable (here α) to average over. For a non-

negative Borel function f on $\mathbb{R}$,

$$\begin{aligned}
\iint f(\omega_x + \alpha) \rho(\omega_x + \alpha) d\alpha \rho(\omega_x) d\omega_x &= \iint f(\omega_x + \alpha) \rho(\omega_x + \alpha) \rho(\omega_x) d\omega_x d\alpha \\
&= \iint f(\omega_x) \rho(\omega_x) \rho(\omega_x - \alpha) d\omega_x d\alpha \\
&= \int f(\omega_x) \rho(\omega_x) \left(\int \rho(\omega_x - \alpha) d\alpha \right) d\omega_x \\
&= \int f(\omega_x) \rho(\omega_x) d\omega_x, \tag{3.17}
\end{aligned}$$

where in the first and third steps we used Fubini, in the second we used translation invariance of Lebesgue measure and in the fourth we used the fact that $\int \rho(\omega_x - \alpha) d\alpha = 1$.

Choose $f(\omega_x) = |G_z(x, y)|^2$; then (3.16) and (3.17) yield

$$\begin{aligned}
|\operatorname{Im} z| \mathbb{E}_x(|G_z(x, y)|^2) &= |\operatorname{Im} z| \mathbb{E}_x \left(\int |G_z^{(\alpha)}(x, y)|^2 \rho(\omega_x + \alpha) d\alpha \right) \\
&\leq \mathbb{E}_x \left(|\operatorname{Im} G_z(x, x)|^{-1} \frac{|G_z(x, y)|^s}{|G_z(x, x)|^s} \int \frac{\rho(\omega_x + \alpha)}{|\alpha + G_z(x, x)^{-1}|^2} d\alpha \right). \tag{3.18}
\end{aligned}$$

We now use Lemma 3.7 below with $w = G_z(x, x)^{-1}$ to conclude

$$|\operatorname{Im} z| \mathbb{E}_x(|G_z(x, y)|^2) \leq C \mathbb{E}_x(|G_z(x, y)|^s),$$

with a constant $C < \infty$ which only depends on $\operatorname{supp} \rho$, but not on x, y and z . $\square$

Lemma 3.7. *Let $0 < s < 1$. There exists a constant $C'_s < \infty$ such that*

$$|\operatorname{Im} w| \cdot |w|^s \int \frac{\rho(\omega_x + \alpha)}{|\alpha + w|^2} d\alpha \leq C'_s$$

uniformly in $w \in \mathbb{C}$ and $\omega_x \sim \mathcal{U}[-1, 1]$.

Proof. The proof is that of [REFERENCE GUNTER STOLZ]. Since $|w|^s \leq |\alpha|^s + |\alpha + w|^s$, we bound:

1.

$$\begin{aligned}
|\operatorname{Im} w| \int \frac{|\alpha|^s \rho(\omega_x + \alpha)}{|\alpha + w|^2} d\alpha &\leq |\operatorname{Im} w| \int \frac{\|\alpha\|^s \rho(\omega_x + \alpha)}{|\alpha + w|^2} d\alpha \\
&\leq |\operatorname{Im} w| \|\alpha\|^s \rho(\omega_x + \alpha) \int \frac{1}{(\alpha + \operatorname{Re} w)^2 + \operatorname{Im} w^2} d\alpha
\end{aligned}$$

for $a, b \in \mathbb{R}$ we have

$$\int \frac{1}{a^2 + b^2} da = 2\pi i \operatorname{Res} \left(\frac{1}{a^2 + b^2}, ib \right) = \frac{2\pi i}{2ib}.$$

Thus for $\rho = \frac{1}{2}\chi_{[-1,1]}$ and $|\alpha| \leq 1 + |\omega| \leq 2$

$$\begin{aligned} |\operatorname{Im} w| \int \frac{|\alpha|^s \rho(\omega_x + \alpha)}{|\alpha + w|^2} d\alpha &\leq \pi \|\alpha\|^s \rho(\omega_x + \alpha) \|_\infty \\ &\leq \pi 2^{s-1}. \end{aligned}$$

2. Now, we bound the second expression.

(a) For $|\operatorname{Im} w|$ big:

Since $|\alpha + w| \geq |\operatorname{Im} w|$ and $\int \rho(\omega_x + \alpha) d\alpha = 1$:

$$I_2 \leq |\operatorname{Im} w| \cdot \frac{1}{|\operatorname{Im} w|^{2-s}} \int \rho(\omega_x + \alpha) d\alpha = \frac{1}{|\operatorname{Im} w|^{1-s}}$$

(b) For $|\operatorname{Im} w|$ small:

$$\begin{aligned} |\operatorname{Im} w| \int \frac{\rho(\omega_x + \alpha)}{|\alpha + w|^{2-s}} d\alpha &\leq |\operatorname{Im} w| \frac{1}{2} \int \frac{d\alpha}{|\alpha + w|^{2-s}} \\ &= |\operatorname{Im} w| \frac{1}{2} \int \frac{1}{|u + \operatorname{Im} w|^{2-s}} du \\ &= |\operatorname{Im} w|^2 \frac{1}{2} \int \frac{1}{|u^2 + \operatorname{Im} w^2|^{\frac{2-s}{2}}} du \\ &= \frac{1}{2} (\operatorname{Im} w)^s \int \frac{1}{|t^2 + 1|^{\frac{2-s}{2}}} dt \end{aligned}$$

by setting $u = \alpha + \operatorname{Re} w$ and $t = \frac{u}{\operatorname{Im} w}$.

To establish the finiteness of the constant, we consider the integral

$$C_s := \int_{-\infty}^{\infty} \frac{1}{(t^2 + 1)^{\frac{2-s}{2}}} dt.$$

For $s \in (0, 1)$, the exponent satisfies $2 - s > 1$, which ensures the asymptotic decay $(t^2 + 1)^{-\frac{2-s}{2}} \sim |t|^{-(2-s)}$ as $|t| \rightarrow \infty$, rendering

the integral absolutely convergent. Then we have

$$|\operatorname{Im} w| \int \frac{\rho(\omega_x + \alpha)}{|\alpha + w|^{2-s}} d\alpha \leq \min \left(\frac{1}{|\operatorname{Im} w|^{1-s}}, C_s \frac{1}{2} |\operatorname{Im} w|^s \right) \leq C'_s 2^{s-1}.$$

□

Now we can proceed with the proof of the theorem.

Proof of Theorem 3.4. For H_ω bounded and self-adjoint, from the spectral theorem, we define the complex Borel spectral measure $\mu_{\delta_x, \delta_y} = \mu_{x,y}$ such that:

$$\mu_{x,y}(B) := \langle \delta_x, \chi_B(H_\omega) \delta_y \rangle.$$

with B Borel in $\mathbb{R}$. The total variation measure $|\mu_{x,y}|$ over the open bounded interval I satisfies

$$\begin{aligned} |\mu_{x,y}|(I) &= \sup_{\substack{g: \mathbb{R} \rightarrow \mathbb{C} \text{ Borel} \\ |g| \leq 1}} \left| \int g(\lambda) d\mu_{x,y}(\lambda) \right| \\ &= \sup_{|g| \leq 1} |\langle \delta_x, g(H_\omega) \chi_I(H_\omega) \delta_y \rangle|, \end{aligned} \quad (3.19)$$

e.g. [Theorem 6.16 Rudin] for $p=1$. This implies:

$$\sup_{t \in \mathbb{R}} |\langle \delta_x, e^{-itH_\omega} \chi_I(H_\omega) \delta_y \rangle| \leq |\mu_{x,y}|(I).$$

From Lusin's theorem, we can extend the bound also for $g \in C_c(I)$:

$$|\mu_{x,y}|(I) = \sup_{\substack{g \in C_c(I) \\ |g| \leq 1}} |\langle \delta_x, g(H_\omega) \delta_y \rangle|.$$

da I bounded open interval. For $g \in C_c(I)$ it follows by elementary analysis (using that g is bounded and uniformly continuous) that, uniformly in $\lambda \in \mathbb{R}$,

$$g(\lambda) = \lim_{\eta \downarrow 0} \frac{\eta}{\pi} \int \frac{g(E)}{(\lambda - E)^2 + \eta^2} dE.$$

By the spectral theorem and dominated convergence, this implies:

$$\begin{aligned}
\langle e_x, g(H_\omega) e_y \rangle &= \int_{\mathbb{R}} g(\lambda) d\mu_{x,y}(\lambda) \\
&= \lim_{\eta \downarrow 0} \int_{\mathbb{R}} \left(\frac{\eta}{\pi} \int_I \frac{g(E)}{(\lambda - E)^2 + \eta^2} dE \right) d\mu_{x,y}(\lambda) \\
&= \lim_{\eta \downarrow 0} \frac{\eta}{\pi} \int_I g(E) \langle \delta_x, (H_\omega - E - i\eta)^{-1} (H_\omega - E + i\eta)^{-1} \delta_y \rangle dE
\end{aligned}$$

Now we can bound $\mathbb{E}(|\mu_{x,y}|)$, considering that $|g(E)| \leq 1$:

$$\begin{aligned}
\mathbb{E}(|\mu_{x,y}|(I)) &\leq \mathbb{E} \left(\lim_{\eta \downarrow 0} \frac{\eta}{\pi} \int_I g(E) \langle \delta_x, (H_\omega - E - i\eta)^{-1} (H_\omega - E + i\eta)^{-1} \delta_y \rangle dE \right) \\
&\leq \mathbb{E} \left(\lim_{\eta \rightarrow 0^+} \frac{\eta}{\pi} \int_I \sum_{z \in \mathbb{Z}^d} |\langle \delta_x, (H_\omega - E - i\eta)^{-1} \delta_z \rangle| |\langle \delta_z, (H_\omega - E + i\eta)^{-1} \delta_y \rangle| dE \right) \\
&\leq \lim_{\eta \rightarrow 0^+} \frac{1}{\pi} \int_I \sum_{z \in \mathbb{Z}^d} (\mathbb{E}(\eta |\langle \delta_x, (H_\omega - E - i\eta)^{-1} \delta_z \rangle|^2))^{1/2} \\
&\quad \cdot (\mathbb{E}(\eta |\langle \delta_z, (H_\omega - E + i\eta)^{-1} \delta_y \rangle|^2))^{1/2} dE,
\end{aligned}$$

where, in this order, Fatou, Fubini and Cauchy–Schwarz (on $\mathbb{E}$) have been used. Now we can use Proposition 3.6 and the hypothesis to obtain:

$$\begin{aligned}
\mathbb{E}(|\mu_{x,y}|(I)) &\leq \frac{C_1}{\pi} \int_I \sum_{z \in \mathbb{Z}^d} (\mathbb{E}|G_\omega(x, z; E + i\eta)|^s)^{1/2} (\mathbb{E}|G_\omega(z, y; E - i\eta)|^s)^{1/2} dE \\
&\leq \frac{C_1 |I| C}{\pi} \sum_{z \in \mathbb{Z}^d} e^{-\frac{\mu}{2}|x-z|} e^{-\frac{\mu}{2}|z-y|}.
\end{aligned}$$

By the triangle inequality, $\frac{\mu}{2}|x-z| + \frac{\mu}{2}|z-y| \geq \frac{\mu}{4}|x-y| + \frac{\mu}{4}|x-z| + \frac{\mu}{4}|z-y|$. Therefore,

$$\mathbb{E}(|\mu_{x,y}|(I)) \leq \left(\frac{C_1 |I| C}{\pi} \sum_{z \in \mathbb{Z}^d} e^{-\frac{\mu}{4}|x-z|} e^{-\frac{\mu}{4}|z-y|} \right) e^{-\frac{\mu}{4}|x-y|} \leq \tilde{C} e^{-\tilde{\mu}|x-y|},$$

with $\tilde{\mu} = \frac{\mu}{4}$ and finite prefactor $\tilde{C} = \frac{C_1 |I| C}{\pi} \sum_{u \in \mathbb{Z}^d} e^{-\frac{\mu}{4}|u|}$. $\square$

From (2.3) we can also easily obtain spectral localisation through Simon–Wolff criterion [book or paper].

Proposition 3.8 (Simon-Wolff criterion). *Let H_ω be a random operator in $l^2(\mathbb{Z}^d)$ with a random potential whose conditional-single site distribution is absolutely continuous. Then if for Lebesgue-almost every $E \in \mathbb{R}$ and $\mathbb{P}$ -almost all ω :*

$$\lim_{\eta \downarrow 0} \sum_{y \in \mathbb{Z}^d} |G_{E+i\eta}(x, y)|^2 < \infty \quad (3.20)$$

H_ω has only pure point spectrum.

Proof can be found in [BOOK]

Theorem 3.9. *Let Theorem 3.1 hold. Then, the spectrum of H_ω consists only of pure point spectrum for a.e. realisation of V_ω .*

The proof, recalls the one proposed in [DISERTORI]

proof of Theorem 4.9. From (3.2) we notice that for all $\lambda > \lambda_{\text{And}}$:

$$\sum_{y \in \mathbb{Z}^d} \mathbb{E}[|G_{E+i\eta}(x, y)|^s] = C \sum_{y \in \mathbb{Z}^d} e^{-\mu|x-y|} < \infty \quad \text{for all } \mu > 0 \quad (3.21)$$

uniformly for $z \in \mathbb{C} \setminus \mathbb{R}$ and $x \in \mathbb{Z}^d$ fixed.

From Proposition 3.6 we notice also that $\sum_{y \in \mathbb{Z}^d} \mathbb{E}[|G_{E+i\eta}(x, y)|^2]$ always exists but might be infinite for $\eta \rightarrow 0$. We argue:

$$\begin{aligned} \mathbb{E} \left[\left(\lim_{\eta \downarrow 0} \sum_{y \in \mathbb{Z}^d} |G_{E+i\eta}(x, y)|^2 \right)^{\frac{s}{2}} \right] &= \mathbb{E} \left[\lim_{\eta \downarrow 0} \left(\sum_{y \in \mathbb{Z}^d} |G_{E+i\eta}(x, y)|^2 \right)^{\frac{s}{2}} \right] \\ &\leq \mathbb{E} \left[\lim_{\eta \downarrow 0} \sum_{y \in \mathbb{Z}^d} |G_{E+i\eta}(x, y)|^s \right] \\ &\leq \liminf_{\eta \downarrow 0} \sum_{y \in \mathbb{Z}^d} \mathbb{E}[|G_{E+i\eta}(x, y)|^s] < \infty, \end{aligned}$$

where in the first two steps we used that the function $x \mapsto x^{\frac{s}{2}}$ is monotone and $(\sum_n a_n)^s \leq \sum_n a_n^s$ for all $a_n \geq 0$ and $0 < s < 1$. The last two inequalities follow by Fatou and (3.21). From Lemma 2.7 we notice that

$$\lim_{\eta \downarrow 0} \sum_{y \in \mathbb{Z}^d} |G_{E+i\eta}(x, y)|^2 < \infty \quad \text{almost surely}$$

□

4 SAW representation

4.1 SAW Definitions

Definition 4.1 (Self-Avoiding Walk). A **self-avoiding walk** on the integer lattice $\mathbb{Z}^d$ of length $N \in \mathbb{N}_0$ is an ordered $(N + 1)$ -tuple of distinct lattice sites

$$\gamma = (\gamma_0, \gamma_1, \dots, \gamma_N) \in (\mathbb{Z}^d)^{N+1}$$

such that consecutive vertices are nearest neighbors, that is,

$$|\gamma_i - \gamma_{i-1}| = 1 \quad \text{for all } i \in \{1, \dots, N\},$$

and $\gamma_i \neq \gamma_j$ for all $i \neq j$.

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Definition 4.2 (Sets of Self-Avoiding Walks). Let $\Lambda \subseteq \mathbb{Z}^d$. For any $x, y \in \Lambda$ and $N \in \mathbb{N}_0$:

- (i) $\mathcal{S}_N^{(\Lambda)}(y, x)$ denotes the set of all self-avoiding walks of length N contained entirely in Λ , starting at $\gamma_0 = x$ and terminating at $\gamma_N = y$. When $\Lambda = \mathbb{Z}^d$, we simply write $\mathcal{S}_N(y, x)$.
- (ii) $\mathcal{S}_N^{(\Lambda)}(x) := \bigcup_{y \in \Lambda} \mathcal{S}_N^{(\Lambda)}(y, x)$ denotes the set of all self-avoiding walks of length N in Λ starting at x .
- (iii) $c_N := |\mathcal{S}_N(x)| = \sum_{y \in \mathbb{Z}^d} |\mathcal{S}_N(y, x)|$ denotes the total number of self-avoiding walks of length N on $\mathbb{Z}^d$ originating from a fixed site (e.g., the origin 0), which is independent of the starting vertex by translation invariance.

Definition 4.3 (Connectivity Constant). The **connective constant** (or connectivity constant) μ_d of the lattice $\mathbb{Z}^d$ is defined as the asymptotic growth rate of c_N :

$$\mu_d := \lim_{N \rightarrow \infty} (c_N)^{1/N}.$$

By subadditivity of $\log c_N$, this limit exists, is positive, and satisfies the strict inequality $\mu_d < 2d - 1$.

Claim 4.4. Let μ_d be the connective constant (or growth rate) of self-avoiding walks on $\mathbb{Z}^d$. Then:

$$\mu_d \leq 2d - 1$$

Proof. Let c_N denote the number of self-avoiding walks (SAWs) of length N starting from the origin $0 \in \mathbb{Z}^d$, and let $S_N(0, x)$ denote the set of such walks ending at $x \in \mathbb{Z}^d$. By definition:

$$c_N = \sum_{x \in \mathbb{Z}^d} |S_N(0, x)| = \sum_{x \in \mathbb{Z}^d} \sum_{\gamma \in S_N(0, x)} 1$$

Consider the generating function (*susceptibility*) with parameter $\xi \in \mathbb{C}$:

$$\sum_{N=0}^{\infty} |\xi|^N c_N = \sum_{N=0}^{\infty} |\xi|^N \sum_{x \in \mathbb{Z}^d} |S_N(0, x)|$$

For $N = 0$, $c_0 = 0$. For $N \geq 1$, a walk cannot immediately step back onto the previous vertex (non-reversing condition). Thus:

- The first step has at most $2d$ choices.
- Each of the subsequent $N - 1$ steps has at most $2d - 1$ choices.

This gives the upper bound:

$$c_N \leq 2d(2d - 1)^{N-1}, \quad \text{for } N \geq 1.$$

Using this upper bound, we can bound the series from above:

$$\begin{aligned} \sum_{N=0}^{\infty} |\xi|^N c_N &\leq \sum_{N=1}^{\infty} |\xi|^N (2d)(2d - 1)^{N-1} \\ &= 2d|\xi| \sum_{k=0}^{\infty} |\xi|^k (2d - 1)^k \quad (\text{setting } k = N - 1) \\ &= 2d|\xi| \sum_{k=0}^{\infty} (|\xi|(2d - 1))^k \end{aligned}$$

This geometric series converges if and only if:

$$|\xi|(2d - 1) < 1 \iff |\xi| < \frac{1}{2d - 1} =: R$$

where $R = \frac{1}{2d-1}$ is the radius of convergence of the bounding series.

By the Cauchy-Hadamard theorem, the radius of convergence R' of the original series $\sum_{N=0}^{\infty} c_N |\xi|^N$ is:

$$R' = \frac{1}{\limsup_{N \rightarrow \infty} c_N^{1/N}} = \frac{1}{\mu_d}$$

Since the original series is bounded term-by-term by a series that converges for $|\xi| < R$, the true radius of convergence must satisfy:

$$R' \geq R \implies \frac{1}{\mu_d} \geq \frac{1}{2d-1}$$

Taking the reciprocal yields:

$$\mu_d \leq 2d - 1.$$

□

Definition 4.5 (SAW Correlation Function). For a parameter $\beta \geq 0$, the **self-avoiding walk correlation function** $C_\beta(y - x)$ between two vertices $x, y \in \mathbb{Z}^d$ is the generating function

$$C_\beta(y - x) := \sum_{N=0}^{\infty} \beta^N |\mathcal{S}_N(y, x)|,$$

defined for all values of β for which the power series is absolutely convergent. By translation invariance, $C_\beta(y - x)$ depends only on the displacement vector $y - x$.

Definition 4.6 (SAW Susceptibility). The **susceptibility** $\chi(\beta)$ of self-avoiding walks is the sum of the two-point correlation function over all end-points:

$$\chi(\beta) := \sum_{x \in \mathbb{Z}^d} C_\beta(x) = \sum_{N=0}^{\infty} c_N \beta^N.$$

The series **converges uniformly absolutely** if and only if $0 \leq |\beta|1/\mu_d$, where $1/\mu_d$ is the critical radius of convergence.

Proposition 4.7 (Exponential Decay of the Correlation Function). *Let $\beta \geq 0$ be such that $\beta < 1/\mu_d$. Then for every $\varepsilon > 0$ there exists a constant $K_\varepsilon < \infty$, independent of x , such that*

$$C_\beta(x) \leq K_\varepsilon ((\mu_d + \varepsilon)\beta)^{|x|} \quad \text{for all } x \in \mathbb{Z}^d.$$

Proof. We know that $|\mathcal{S}_N(0, x)| = 0$ whenever $N < |x|$, and the sum defining $C_\beta(x)$ may be restricted to $N \geq |x|$:

$$C_\beta(x) = \sum_{N=|x|}^{\infty} \beta^N |\mathcal{S}_N(x, 0)| \leq \sum_{N=|x|}^{\infty} \beta^N c_N. \quad (4.1)$$

Fix $\varepsilon > 0$. Since $\mu_d = \lim_{N \rightarrow \infty} c_N^{1/N}$, by the definition of limit there exists $N_0 \in \mathbb{N}$ such that

$$c_N^{1/N} < \mu_d + \varepsilon \quad \text{for all } N \geq N_0,$$

that is,

$$c_N < (\mu_d + \varepsilon)^N \quad \text{for all } N \geq N_0.$$

For $N = 0, 1, \dots, N_0 - 1$, define

$$K_\varepsilon := \max \left(1, \max_{0 \leq N < N_0} \frac{c_N}{(\mu_d + \varepsilon)^N} \right),$$

which is finite, being the maximum of finitely many finite quantities. By construction it holds:

$$c_N \leq K_\varepsilon (\mu_d + \varepsilon)^N \quad \text{for all } N \geq 0.$$

Combining it with (4.1) we obtain

$$C_\beta(x) \leq K_\varepsilon \sum_{N=|x|}^{\infty} [(\mu_d + \varepsilon)\beta]^N.$$

Choosing ε small enough that $r := (\mu_d + \varepsilon)\beta < 1$ (possible since $\beta < 1/\mu_d$), the tail of the geometric series equals

$$\sum_{N=|x|}^{\infty} r^N = \frac{r^{|x|}}{1 - r}.$$

Hence

$$C_\beta(x) \leq \frac{K_\varepsilon}{1 - (\mu_d + \varepsilon)\beta} [(\mu_d + \varepsilon)\beta]^{|x|}.$$

where $\frac{K_\varepsilon}{1 - (\mu_d + \varepsilon)\beta}$ is still finite and x -independent. □

4.2 SAW representation of $\mathbf{G}$

The proof of the theorem 3.1 is similar to the one given by Schenker in [1]. We are going to work with the Green's function defined on a volume where one site is removed every stage. Therefore, for any $\Lambda \supset \mathbb{Z}^d$ (finite or infinite)

the Green function changes:

$$G^\Lambda(x, y; z) := \begin{cases} (H_\omega^\Lambda - z), & \forall x, y \in \Lambda \\ 0 & , \text{otherwise} \end{cases} \quad (4.2)$$

where $H_\omega^\Lambda - z \in \mathbb{C}^{\Lambda \times \Lambda}$ is the matrix obtained by restricting $H_\omega - z$ to Λ . In this thesis we will consider $\Lambda = \mathbb{Z}^d$, but we leave the notation Λ to highlight the fact that the same result holds for any volume.

To notice is also that $H_\omega^\Lambda - z$ is still invertible for all $z \in \mathbb{C} \setminus \mathbb{R}$. In this section, for simplicity during calculations, we will write $G^\Lambda(x, y)$ instead of $G^\Lambda(x, y; z)$.

Theorem 4.8. *Let $x, y \in \mathbb{Z}^d$, $x \neq y$, $\Lambda \subseteq \mathbb{Z}^d$. G^Λ is defined as in (4.1). It holds*

$$G^\Lambda(x, y) = \sum_{n=1}^N \sum_{\gamma \in \mathcal{S}_n(x, y)} \prod_{i=1}^n G^\Lambda(x, x) \cdot G^{\Lambda_i}(\gamma_i, \gamma_i) + R_N \quad (4.3)$$

where $\Lambda_i = \Lambda \setminus \{x, \gamma_1, \dots, \gamma_{i-1}\}$ and

$$R_N = \sum_{x \sim y} \sum_{\gamma \in \mathcal{S}_N(0, x)} \prod_{i=1}^{N-1} G^\Lambda(x, x) \cdot G^{\Lambda_i}(\gamma_i, \gamma_i) \cdot G^{\Lambda_N}(\gamma_N, y). \quad (4.4)$$

Claim 4.9.

$$G^\Lambda(x, y) = G^\Lambda(x, x) \sum_{x_1 \sim x} G^{\Lambda_1}(x_1, y) \quad (4.5)$$

Proof. Using the resolvent identity for $A = (H_\omega^\Lambda - z)$ and $B = (H_\omega^{\{x\}} \oplus H_\omega^{\Lambda \setminus \{x\}} - z)$, we obtain:

$$\begin{aligned} & (H_\omega^\Lambda - z)^{-1} - (H_\omega^{\{x\}} \oplus H_\omega^{\Lambda \setminus \{x\}} - z)^{-1} \\ &= -(H_\omega^\Lambda - z)^{-1} \left(H_\omega^\Lambda - (H_\omega^{\{x\}} \oplus H_\omega^{\Lambda \setminus \{x\}}) \right) (H_\omega^{\{x\}} \oplus H_\omega^{\Lambda \setminus \{x\}} - z)^{-1} \end{aligned}$$

We know:

$$\begin{aligned} H_\omega^\Lambda &= 2d - \sum_{x_1 \sim y} \delta_{x_1} + \lambda V_\omega \quad \forall y \in \Lambda \\ H_\omega^{\{x\}} &= \lambda \omega_x + 2d \\ H_\omega^{\Lambda \setminus \{x\}} &= 2d - \sum_{x_1 \sim y} \delta_{x_1} + \lambda V_\omega - \lambda \omega_x \quad \forall y \in \Lambda \setminus \{x\} \end{aligned}$$

$$\implies H_\omega^\Lambda - (H_\omega^{\{x\}} \oplus H_\omega^{\Lambda \setminus \{x\}}) = - \sum_{x_1 \sim x} \delta_{x_1} =: T_{x, x_1}$$

Now considering the matrix elements in (x, y) with $x \neq y$ and matrix multiplication:

$$(H_\omega^\Lambda - z)_{x, y}^{-1} - 0 = - \sum_{x_1, x_2} (H_\omega^\Lambda - z)_{x, x_1}^{-1} T_{x_1, x_2} (H_\omega^{\{x\}} \oplus H_\omega^{\Lambda \setminus \{x\}} - z)_{x_2, y}^{-1}$$

For $T_{x_1, x_2} \neq 0$, we need $x_1 = x$ and $x_2 \sim x_1$.

We obtain:

$$G^\Lambda(x, y) = \sum_{x_1 \sim x} G^\Lambda(x, x) G^{\Lambda \setminus \{x\}}(x_1, y)$$

□

Proof of Theorem 5.8. Now we separate the case in which $x_1 = y$ (which implies $|x - y| = 1$).

$$G^\Lambda(x, y) = G^\Lambda(x, x) \left(\sum_{x_1 \sim x, x_1 \neq y} G^{\Lambda_1}(x_1, y) + I_{|x-y|=1} G^{\Lambda_1}(y, y) \right)$$

Iterating N times, it gives (4.3).

□

4.3 Proof with SAW

Now we use self-avoiding-walks to prove the theorem (3.1).

Proof. We take the average of (4.2), and using $|\sum a|^s \leq \sum |a|^s$ (which follows from triangle inequality), we have:

$$\mathbb{E} [|G^\Lambda(x, y)|^s] \leq \sum_{n=1}^N \sum_{\gamma \in \mathcal{S}_n(x, y)} \mathbb{E} \left[|G^\Lambda(x, x)|^s \prod_{i=1}^n |G^{\Lambda_i}(\gamma_i, \gamma_i)|^s \right] + \quad (4.6)$$

$$\sum_{x \sim y} \sum_{\gamma \in \mathcal{S}_N(0, x)} \mathbb{E} [|G^\Lambda(x, x)|^s \prod_{i=1}^{N-1} |G^{\Lambda_i}(\gamma_i, \gamma_i)|^s \cdot |G^{\Lambda_N}(\gamma_N, y)|^s] \quad (4.7)$$

The resolvent $G^{\Lambda_j}(w, w')$ does not depend on the random variables ω_{x_i} , $i = 0, \dots, j-1$. Hence, recursive applications of the a priori bound (3.2) yield

$$\mathbb{E} \left[|G^\Lambda(x, x)|^s \prod_{j=1}^n |G^{\Lambda_j}(\gamma_j, \gamma_j)|^s \right] \leq \left(\frac{1}{1-s} \frac{1}{\lambda^s} \right)^{n+1}$$

and

$$\begin{aligned} & \mathbb{E} \left[|G^\Lambda(x, x)|^s \prod_{j=1}^{N-1} |G^{\Lambda_j}(\gamma_j, \gamma_j)|^s \cdot |G^{\Lambda_{N-1}}(\gamma_N, y)|^s \right] \\ & \leq \left(\frac{1}{1-s} \cdot \frac{1}{\lambda^s} \right)^N \cdot \mathbb{E} [|G^{\Lambda_{N-1}}(\gamma_N, y)|^s] \leq \left(\frac{1}{1-s} \frac{1}{\lambda^s} \right)^N \frac{1}{|\operatorname{Im}(z)|^s} \end{aligned}$$

Where we used a priori bound and lemma 3.5.

Inserting these estimates in the sums above, we get:

$$\mathbb{E} \left(|G^{(\Lambda)}(x, y)|^s \right) \leq \sum_{n=0}^N \beta(s)^{1+n} |\mathcal{S}_n^{(\Lambda)}(y, x)| + \operatorname{Err}(N) \frac{1}{|\operatorname{Im} z|^s} \quad (4.8)$$

with $\operatorname{Err}(N) := \beta(s)^N |\mathcal{S}_N^{(\Lambda)}(x)|$ $\beta(s) = \frac{1}{1-s} \frac{1}{\lambda^s}$ Let's consider the susceptibility $\chi(\beta)$ of the self avoiding walks in $\mathcal{S}_N^{(\Lambda)}(x)$:

$$\chi(\beta) = \sum_{x \in \mathbb{Z}^d} \sum_{N=0}^{\infty} \beta(s)^N |\mathcal{S}_N^{(\Lambda)}(x)| = \sum_{N=0}^{\infty} c_N \beta(s)^N. \quad (4.9)$$

The series converges absolutely if and only if $0 \leq |\beta| < 1/\mu_d$, where $1/\mu_d = \lim_{N \rightarrow \infty} c_N^{1/N}$ is the critical radius of convergence. Thus, for $|\beta(s)| = \frac{1}{|(1-s)\lambda^s|} \leq \frac{1}{\mu_d}$, $\chi(\beta)$ converges uniformly, and most importantly:

$$\operatorname{Err}(N) \longrightarrow 0, \quad \text{for } N \rightarrow \infty$$

We send N to the infinity and obtain from (4.7):

$$\mathbb{E} \left(|G^{(\Lambda)}(x, y)|^s \right) \leq \sum_{n=0}^{\infty} \beta(s)^{1+n} |\mathcal{S}_n^{(\Lambda)}(y, x)| \quad (4.10)$$

Taking the supremum over volumes Λ now yields

$$\sup_{\Lambda \subset \mathbb{Z}^d} \mathbb{E} \left(|G^{(\Lambda)}(x, y)|^s \right) \leq \beta(s) C_{\beta(s)}(x - y). \quad (4.3)$$

It remains to establish for which λ we may find $s \in (0, 1)$ with $\beta(s) < 1/\mu_d$. As above, for $\lambda > e$ the unique critical point is $s_{\text{crit}} = 1 - 1/\ln \lambda$ and

$$\min_{s \in (0, 1)} \beta(s) = \beta(s_{\text{crit}}) = \frac{e \ln \lambda}{\lambda}$$

which is less than $1/\mu_d$ if and only if $\lambda > \lambda_{\text{And}}$. Finally we get:

$$\mathbb{E} \left(|G^{(\Lambda)}(x, y; z)|^{1 - \frac{1}{\ln \lambda}} \right) \leq \beta(\lambda) K_\epsilon \ln \lambda e^{-m_\epsilon(\lambda)|x-y|} \quad (4.11)$$

where

$$m_\epsilon(\lambda) = -\ln \beta(\lambda) - \ln(\mu_d + \epsilon) > 0.$$

□

5 Connection to Long-Range SAW

The purpose of this chapter is to extend the self-avoiding walk representation of the resolvent, to random Hamiltonians with long-range hopping. In particular, we compare this setting with the previous model and apply the fractional moment method to study localization. The difference lies in the kinetic operator, which is replaced by the fractional discrete Laplacian $(-\Delta)^\alpha$ with $0 < \alpha < 1$. This change leads to a polynomial decay bound for the averaged Green's function, in sharp contrast with the standard exponential decay. This chapter is based on the work of M. Disertori, R.M. Escobar and C. Rojas-Molina in 2024 [1].

5.1 The fractional Laplacian

For the long-range, the kinetic operator also connects arbitrarily distant sites in the lattice, rather than just the first neighbors.

Definition 5.1. Let $-\Delta$ be defined as in 1.4 in Section 1. The fractional discrete Laplacian is $(-\Delta)^\alpha$ for $0 < \alpha < 1$

Just as the standard discrete Laplacian, $(-\Delta)^\alpha$ is bounded, self-adjoint and translation invariant. Furthermore, it satisfies

$$(-\Delta)_{x,x}^\alpha > 0 \quad \forall x \in \mathbb{Z}^d, \quad (-\Delta)_{x,y}^\alpha \leq 0 \quad \forall x \neq y \text{ in } \mathbb{Z}^d$$

Moreover, its off-diagonal matrix elements exhibit polynomial decay: there exist constants $0 < c_{\alpha,d} \leq C_{\alpha,d} < \infty$ such that

$$\frac{c_{\alpha,d}}{|x-y|^{d+2\alpha}} \leq -(-\Delta)_{x,y}^\alpha \leq \frac{C_{\alpha,d}}{|x-y|^{d+2\alpha}} \quad (5.1)$$

From this bound, one deduces the summability of fractional powers of the matrix elements:

Lemma 5.2. For any $\alpha \in (0, 1)$ and $s \in \left(\frac{d}{d+2\alpha}, 1\right]$,

$$\sum_{x \in \mathbb{Z}^d} |(-\Delta)_{0,x}^\alpha|^s < \infty \quad (5.2)$$

Proof. From (5.1):

$$\sum_{|x| \geq 1, x \in \mathbb{Z}^d} |(-\Delta)_{0,x}^\alpha|^s \leq C_{\alpha,d}^s \sum_{|x| \geq 1, x \in \mathbb{Z}^d} \frac{1}{|x|^{(d+2\alpha)s}}.$$

By standard comparison of the lattice sum with an integral on $\mathbb{R}^d$:

$$\sum_{|x| \geq 1, x \in \mathbb{Z}^d} \frac{1}{|x|^p} \leq C_d \int_{|z| > 1} \frac{1}{|z|^p} dz.$$

Passing to spherical coordinates in $\mathbb{R}^d$, the integral reads

$$\begin{aligned} \int_{|z| > 1} \frac{1}{|z|^p} dz &= |\mathbb{S}^{d-1}| \int_1^\infty \frac{r^{d-1}}{r^p} dr \\ &= |\mathbb{S}^{d-1}| \int_1^\infty r^{d-1-p} dr, \end{aligned}$$

where $|\mathbb{S}^{d-1}| = \frac{2\pi^{d/2}}{\Gamma(d/2)}$ denotes the surface area of the unit sphere in $\mathbb{R}^d$.

The radial integral converges if and only if the power satisfies

$$d - 1 - p < -1 \iff p > d.$$

Substituting $p = s(d + 2\alpha)$, this condition becomes

$$s(d + 2\alpha) > d \iff s > \frac{d}{d + 2\alpha}.$$

Since $\alpha > 0$ implies $d + 2\alpha > d$; for every $s \in \left(\frac{d}{d+2\alpha}, 1\right]$, we conclude that

$$\sum_{x \neq 0} |(-\Delta)_{0,x}^\alpha|^s \leq C_{\alpha,d}^s \sum_{x \neq 0} \frac{1}{|x|^{s(d+2\alpha)}} < \infty.$$

□

5.2 The SAW Representation for Long-Range Resolvents

We consider the fractional Anderson Hamiltonian

$$H_{\alpha,\omega} = (-\Delta)^\alpha + \lambda V_\omega \quad \text{on } \ell^2(\mathbb{Z}^d), \quad (5.3)$$

where $0 < \alpha < 1$ and V_ω is a diagonal matrix with i.i.d. random entries. In this case, we generalise directly for an arbitrary continuous distribution with a compactly supported, bounded density.

The Green's function $G_z = (H_{\alpha,\omega} - z)^{-1}$ is well-defined for all $z \in \mathbb{C} \setminus \mathbb{R}$, and its decay at strong disorder is closely related to the polynomial decay of the fractional Laplacian.

As proven in [Theorem 1 and 2 of PAPER DISERTORI], the averaged fractional Green's function satisfies

$$\mathbb{E}[|G_z(x, y)|^s] \leq \frac{K_0 \theta_s}{\lambda^s} \frac{1}{|x - y|^{d+2\alpha_s}} \quad \forall x, y \in \mathbb{Z}^d, x \neq y, \quad (5.4)$$

uniformly in $z \in \mathbb{C} \setminus \mathbb{R}$, for all $\lambda > \lambda_0(s) := \left(\frac{\theta_s}{R_{\chi_D^{\text{SAW}}}^s} \right)^{\frac{1}{s}}$. Here, $R_{\chi_D^{\text{SAW}}}$ is the radius of convergence of the susceptibility, and $2\alpha_s := s(d + 2\alpha) - d$. The constant $K_0 = K_0(d, \alpha, s)$ comes from the two-point function bound of the self-avoiding walk, and $\theta_s = \frac{1}{1-s} M_1(P_0)^s$ comes from the *a priori bound* evaluated at $\tau = 1$.

In the long-range scenario, we use the geometric resolvent expansion to represent the fractional moments as a sum over self-avoiding paths. Because $(-\Delta)^\alpha$ possesses long-range jumps, we are not just considering nearest neighbors, but arbitrary sites on the lattice. We define the step weight kernel for the SAW as $D(x, y) := |(-\Delta)^\alpha(x, y)|^s$, which generates a transition probability $p(x, y) := \frac{D(x, y)}{\sum_{z \neq x} D(x, z)}$. This is well-defined for $s \in \left(\frac{d}{d+2\alpha}, 1 \right)$ because the kernel is summable: $\sum_{x \neq 0} |(-\Delta)^\alpha(0, x)|^s < \infty$ (See (5.2)).

As in the previous Chapter, in order to prove (5.4), we organize the proof into three main steps:

1. **Resolvent Identity Iteration:** We expand the restricted Green's function $G_z^\Lambda(x, y)$ by iterating the resolvent identity N times, pulling out the off-diagonal elements $\Delta^\alpha(\gamma_j, \gamma_{j+1}) = -(-\Delta)^\alpha(\gamma_j, \gamma_{j+1}) > 0$ along self-avoiding paths $\gamma \in \mathcal{S}_n(x, y)$. Since $0 < \alpha < 1$, we keep the Laplacian fractional instead of reducing it to nearest-neighbor hopping.
2. **Averaging and A Priori Bound:** We take the fractional moment

of the expansion and apply subadditivity. Using the a priori bound to decouple the diagonal resolvent terms, we obtain:

$$\mathbb{E}[|G_z^\Lambda(x, y)|^s] \leq \frac{\theta_s}{\lambda^s} C_\beta^D(y - x) + \frac{1}{|\operatorname{Im} z|^s} \operatorname{Err}(N), \quad (5.5)$$

where C_β^D is the SAW two-point function with fugacity $\beta = \frac{\theta_s}{\lambda^s}$. For $\lambda > \lambda_0(s)$, we have $\beta < R_{\chi_\beta^D}$, which ensures the susceptibility is finite and $\operatorname{Err}(N) \rightarrow 0$ as $N \rightarrow \infty$.

3. **Two-Point Function Decay:** Finally, by [Lemma 4 of PAPER DISERTORI], the SAW correlation function is bounded by the generating kernel's decay:

$$C_\beta^D(y - x) \leq K_0 \frac{1}{|y - x|^{d+2\alpha_s}}, \quad (5.6)$$

yielding the final polynomial bound (5.4).

5.3 The Localisation Criteria

A natural question that may arise is how localisation is proven without the exponential decay of the fractional moments of the Green's function. Due to the polynomial tail of the fractional Laplacian, exponential decay is not expected; instead, it is only possible to prove a weaker version of dynamical localisation where only moments $\beta < 2\alpha$ of the position operator are finite. Spectral localisation is proven via the Simon-Wolff theorem.

Lemma 5.3. *Let (5.4) hold uniformly in $z \in \mathbb{C} \setminus \mathbb{R}$ and for $\lambda > \lambda_0(s)$. It follows that:*

- (i) *The spectrum of $H_{\alpha, \omega}$ consists only of pure point spectrum for a.e. $\omega \in \Omega$.*
- (ii) *For a.e. $\omega \in \Omega$ and for all E in the almost sure spectrum, the corresponding normalized eigenfunction φ_E satisfies:*

$$|\varphi_E(y, \omega)|^2 \leq \frac{C_t(\omega)}{(1 + |y - x_E(\omega)|)^t} \quad \forall 0 < t < 2\alpha_s, \quad (5.7)$$

where $x_E(\omega)$ is a localization center and $C_t(\omega)$ is an integrable random variable.

(iii) For $0 < \beta < 2\alpha_s$, we have for a.e. $\omega \in \Omega$

$$\sup_{t \in \mathbb{R}} \sum_{x \neq 0} |x|^\beta |e^{-itH_{\alpha,\omega}}(0, x)|^2 < \infty. \quad (5.8)$$

In particular, if $\alpha > \frac{1}{2}$, the first moment of the wave function initially localized at the origin and evolving under the action of $H_{\alpha,\omega}$ is finite.

Proof. (i) From (5.4), we verify the summability of the Green's function moments:

$$\sum_{x \in \mathbb{Z}^d} \mathbb{E}[|G_z(0, x)|^s |x|^t] \leq \frac{K_0 \theta_s}{\lambda^s} \sum_{x \neq 0} \frac{1}{|x|^{d+2\alpha_s-t}} < \infty \quad \forall t \in [0, 2\alpha_s), \quad (5.9)$$

uniformly in $z \in \mathbb{C} \setminus \mathbb{R}$ and for all $\lambda > \lambda_0(s)$. Setting $t = 0$ and applying Fatou's lemma ensures that $\liminf_{\eta \downarrow 0} \sum_y |G_{E+i\eta}(x, y)|^2 < \infty$ a.s.. We then apply the Simon-Wolff criterion to obtain spectral localisation almost everywhere.

- (ii) The eigenfunction polynomial decay follows from the summability bound (5.9) and standard fractional moment arguments.
- (iii) The dynamical bounds follow from bounding the eigenfunction correlator $Q(x, 0, \Sigma)$, which is majorized by the integrated Green's function moments. Since $\mathbb{E}[Q(x, 0, \Sigma)] \leq \frac{C}{|x|^{d+2\alpha_s}}$, the sum $\sum_{x \neq 0} |x|^\beta \mathbb{E}[Q(x, 0, \Sigma)]$ converges for $\beta < 2\alpha_s$, yielding the result.

□